CROSCIM: a multiscale, observation-conditioned generative model for sea ice

from native observations to scale-interaction experiments · levels $\mathcal{C}$ 32 km, $\mathcal{M}$ 4 km, $\mathcal{H}$ 1 km (nested polar-stereographic grid)

1 Data & dataloader

Native observation tables (Zarr + time index)

- PMW SIC, PMW drift, CS2SMOS SIT
- CIMR SIC (~5 km), SAR drift
- ASIP SAR SIC (1 km), IR IST, CRISTAL

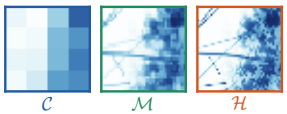
level $\ell(n) = \arg\min_{\ell} |\log r_n - \log \Delta_\ell|$

ERA5 forcing, dense on $\mathcal{C}$: T_{2m}, q_{2m}, u_{10}, v_{10}, SW ↓ , LW ↓ , snowfall

Per sample $(t_0,$ window$)$

1. select obs in $[t_0 - L, t_0]$ (and $(t, t+\Delta]$) by time index, no interpolation
2. split input / target (stream, region or time; deterministic if precomputed)
3. inputs → tokens per level: dense streams as patches + masks, irregular as packed ragged arrays
4. targets → binned per level (R on obs) + masks m_ℓ: consistent hierarchy
5. batch: $\mathcal{C}$ full domain, $\mathcal{M}$ window + halo, $\mathcal{H}$ window in the ROI
6. $\Pi_\uparrow z_\mathcal{H}$ outside the window: precomputed or no-grad

2 Model: cascade of conditional generators



$Y = (X_\mathcal{C}, \delta X_\mathcal{M}, \delta X_\mathcal{H}),\ R\delta X = 0$

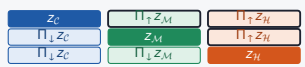
$X_\mathcal{H} = PX_\mathcal{C} + P\delta X_\mathcal{M} + \delta X_\mathcal{H}$

R, P fixed; no learned decoder

Observation latents and cross-scale sharing

$z_\ell = E_\ell(O_\ell)$: learned, reduced analogue of $H^\top R^{-1} y$

$\Pi_\uparrow$: attention pooling over children (fan-in 16/64); $\Pi_\downarrow$: broadcast

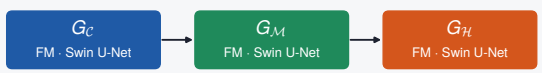


+ F (all levels)
+ state s_ℓ^t (Stage 2)
bold frame: bottom-up

Chain rule = posterior, as in data assimilation

$p(Y \mid O) = p_\theta(X_\mathcal{C} \mid c_\mathcal{C})\, p_\theta(\delta X_\mathcal{M} \mid X_\mathcal{C}, c_\mathcal{M})\, p_\theta(\delta X_\mathcal{H} \mid X_\mathcal{C}, \delta X_\mathcal{M}, c_\mathcal{H})$

exact iff each $c_\ell = g_\ell(O)$ is sufficient; FM: $\mathcal{L}^*(g) - \mathcal{L}^*(O) = \mathbb{E}\|v_O^* - v_g^*\|^2$



one sample per member, $\mathcal{C} \to \mathcal{M} \to \mathcal{H}$; for δ levels: $v \leftarrow v - PRv$, bounds by censoring on the recomposed field

Stage 1 $p(Y_{t_0} | z(O_{t_0-L:t_0}), F)$ Stage 2 $p(Y^{t+\Delta} | Y^t, F, [z(O)])$

3 Training

Per level, conditional flow matching

$x_s = s\,x_1 + (1-s)\,\varepsilon$

$\mathcal{L} = \Sigma_\ell \mathbb{E}[m_\ell \odot (\frac{1}{2} r_\ell^2 + \log \sigma_\ell)]$

- $\varepsilon \leftarrow \varepsilon - PR\varepsilon$ for δ levels
- coarse conditioning $+\eta$ (truth → samples)
- levels trained in parallel

Dropout by group

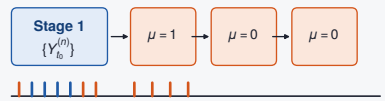
- obs groups (μ) and $\Pi_\uparrow$ groups → ∅
- makes ablations, closure-off and prior / posterior in-distribution

Order and stability

1. Stage 1 on observation windows
2. archive of analysis ensembles
3. Stage 2 on those analyses: pushforward 2–4 steps, obs dropout, memory σ_ℓ
4. held-out streams for verification

4 Inference & rollout

Analysis then rollout (physical state)



$\mu = 1$: re-conditioning by obs (learned filter)
$\mu = 0$: forecast with forcing $F_{t:t+\Delta}$

Small scales without obs

memory: $x_0^\ell = Y_\ell^t + \sigma_\ell \varepsilon$

- $t < \tau_\ell$: prognostic (carried, advected)
- $t > \tau_\ell$: conditional climatology $p(\delta X_\mathcal{H} | X_\mathcal{C}, \delta X_\mathcal{M}, F)$

Two modes, one model

- short range, ROI: $\mathcal{H}$ every step, closure $\Pi_\uparrow s_\mathcal{H}^t$
- long range / outside ROI: $\mathcal{C}$ carries sub-grid moments $\phi_\mathcal{C}$ (ITD-like)
- $\mathcal{C}$, $\mathcal{M}$ full domain; $\mathcal{H}$ chunks in ROI; latents encoded once per step

5 Experiments & diagnostics

Scale interactions (paired noise)

- small→large: $\delta X'_\mathcal{H} = \Pi_0 T(\delta X_\mathcal{H})$, R preserved
- T: smooth, swap details, ×α, regional
- $I_\uparrow(\tau) = \|\overline{X_\mathcal{C}^{int} - X_\mathcal{C}^{ctrl}}\| / \sigma_\mathcal{C}$
- closure off: $\Pi_\uparrow \to \varnothing$ in $c_\mathcal{C}$ each step
- large→small: $\Delta X_\mathcal{C}$ → fine statistics $I_\downarrow$
- dynamical (Y^t) vs observational ($O_\mathcal{H}$) path

Data-assimilation diagnostics

- prior ($\mu = 0$) vs posterior ($\mu = 1$)
- increments, spread reduction
- per-stream impact (OSE by dropout)
- Desroziers, rank histograms,
- $(y - H\hat{X})/\sigma$ on held-out streams
- explicit-H guidance for new sensors

Skill and physics

- CRPS, spread–skill per level and lead
- PSD per level; effective resolution λ
- predictability τ_ℓ (spread saturation)
- $p(\delta X_\mathcal{H} | X_\mathcal{C})$ stable in rollouts
- volume and bounds (SIC×SIT, SIC $\in [0, 1]$)
- SIC, SIT, drift, snow, IST jointly

Ablations and baselines

- current CROSCIM (deterministic cascade)
- cascade without $\Pi_\uparrow$ (top-down only)
- placebo summaries (same statistics)
- summary size K, analytic vs learned
- memory σ_ℓ; Markov vs multi-step
- persistence, single-resolution FM